

XV.—On the Estimation of Statistical Parameters. By **A. C. Aitken**, D.Sc., F.R.S., Mathematical Institute, University of Edinburgh, and **H. Silverstone**, Ph.D., New Zealand.

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I. INTRODUCTORY.

THE present paper communicates some of the results given in a thesis submitted by Mr Silverstone to the University of Edinburgh in May 1939. The starting-point, the adoption of the postulates of unbiased estimate and minimum sampling variance, and the analogies which would emerge with the theories of maximum likelihood and of linear estimation by Least Squares, were suggested by the senior author. Thanks are due to Dr R. P. Gillespie for some helpful preliminary discussion in 1937 and 1938.

The senior author is also responsible for the composition of this paper and for modifications of the original material. Owing to the hazards and delays of correspondence arising from the present war, it has not been possible for the authors, one living in Scotland and the other in New Zealand, to continue their collaboration.

2. MAXIMUM PROBABILITY DENSITY AND MINIMUM VARIANCE.

In the theory of linear estimation by Least Squares from linear observational equations affected by error, the normal equations for the optimal values of the unknowns can be deduced by two quite different sets of postulates. The first, (A), proceeds by (i) assuming a normal distribution of errors of observation, and (ii) accepting as optimal values of the unknowns those which make the compound probability density a maximum; the second, (B), proceeds by (i) assuming that the optimal values are unbiased linear combinations of the observations, and (ii) accepting those particular linear combinations for which the compound variance of error is a minimum. It is to be noted that in (B) the law of error of the observations is not restricted to be of normal or of any assigned type, and could indeed be different for each observation.

In the theory of estimation of statistical parameters the principle of maximum likelihood (Fisher, 1921) bears a certain resemblance to postulate

(ii) of the first set (A) of postulates of Least Squares. The principle accepts, in fact, as best estimate of an unknown parameter θ in a probability function $\phi(x; \theta)$ of assigned type, that function $t(x_1, x_2, \dots, x_n)$ of the sample values x_i which makes the compound probability density of the x_i a maximum. The vindication of the principle is largely *a posteriori*, arising from the circumstance that if the estimating function t has a sampling distribution of normal type, then, within the class of unbiased analytic functions estimating θ , t has also the least possible variance. Even when the distribution of t is not normal, it tends more and more to become so as the size of sample is increased, and the property of minimum variance tends to hold with closer and closer approximation.

This analogy with the position in Least Squares suggests a basis for estimation alternative to that of maximum likelihood and closely resembling the postulates (B). Let us adopt postulates (i) of unbiased or consistent estimate, (ii) of minimum variance of estimating function. The investigation of the consequences of these postulates sets the principle of maximum likelihood in an interesting light. For example, simple conditions emerge under which maximum likelihood provides an estimate accurately possessing minimum variance, even though the sample is finite and the distribution of the estimating function is not normal. At the same time the actual value of the minimum variance is obtained with special ease.

The arbitrariness of the postulates needs no emphasis. They are inevitably arbitrary; but so are those of any proposed alternative set. The restriction to a linear operation in the condition of consistency is admittedly a severe one. One is free to choose a condition of consistency different from the one here adopted, but the subsequent analysis will be found decidedly less tractable, perhaps impossible. Again, since, as is well known, many plausibly posed questions in the Calculus of Variations do not admit of analytic solution, it evokes no surprise that wide classes of probability functions are not amenable to estimation of parameters by the suggested postulates.

3. POSTULATES FOR ESTIMATION OF PARAMETERS.

Let $\phi(x; \theta)$ be a probability function containing an unknown parameter θ , which is to be estimated from a sample $x_1, x_2, \dots, x_n$ of n values of x . The range of x is presumed to be independent of θ ; and ϕ is presumed to be uniformly differentiable with respect to θ . Let us denote the compound probability density of the sample by

$$\Phi(x_1, x_2, \dots, x_n; \theta) \equiv \Phi.$$

It is required to estimate θ by a function $t(x_1, x_2, \dots, x_n) \equiv t$, where t is independent of θ . Let us adopt the two following postulates:—

(i) That t shall be an unbiased or consistent estimate of θ , in the sense that the mean value or expectation of t over all n -ary samples $x_1, x_2, \dots, x_n$ shall be θ ; that is,

$$\iiint \dots t \Phi dx_1 dx_2 \dots dx_n = \theta. \quad (1)$$

(ii) That the sampling variance of t over all such samples shall be a minimum; that is,

$$\iiint \dots (t - \theta)^2 \Phi dx_1 dx_2 \dots dx_n = \text{minimum}. \quad (2)$$

We have also the necessary condition of total probability, namely,

$$(iii) \quad \iiint \dots \Phi dx_1 dx_2 \dots dx_n = 1. \quad (3)$$

The problem is thus a minimal problem in the Calculus of Variations, of positive definite type and formally simple. Let us write the Euler equation for its solution in the form

$$(t - \theta)\Phi - \lambda(\theta)\Phi_\theta = 0, \quad \text{where} \quad \Phi_\theta = \frac{\partial}{\partial \theta} \Phi, \quad (4)$$

the Lagrange function $\lambda(\theta)$ being independent of $x_1, x_2, \dots, x_n$. We then have

$$\begin{aligned} t &= \theta + \lambda \Phi_\theta / \Phi \\ &= \theta + \lambda \left(\frac{\partial}{\partial \theta} \log \Phi \right), \end{aligned} \quad (5)$$

where t must be independent of θ . In short, an estimating function t exists, provided that the derivative of $\log \Phi$ with respect to θ can be resolved as follows:

$$\frac{\partial}{\partial \theta} \log \Phi = (t - \theta) / \lambda(\theta). \quad (6)$$

Two standard examples will serve to illustrate

Example 1.—To estimate μ in $\phi(x; \mu) \equiv (2\pi)^{-\frac{1}{2}} \exp -\frac{1}{2}(x - \mu)^2$.

Here

$$\begin{aligned} \Phi &= (2\pi)^{-\frac{1}{2}n} \exp \left\{ -\frac{1}{2} \sum (x_j - \mu)^2 \right\}, \\ \frac{\partial}{\partial \mu} \log \Phi &= \sum x_j - n\mu \\ &= (t - \mu) / \lambda(\mu) \end{aligned}$$

provided that $\lambda(\mu) = 1/n$, in which case $t = \sum x_j / n = \bar{x}$. Thus the estimate of μ is $\bar{x}$, the mean of the observations. As for λ , let us anticipate the general result by noting that its value here, $1/n$, is the minimum variance in question, namely, the sampling variance of $\bar{x}$.

Example 2.—To estimate σ^2 in $(2\pi\sigma^2)^{-\frac{1}{2}} \exp -\frac{1}{2}x^2/\sigma^2$.

Here

$$\begin{aligned}\frac{\partial}{\partial \sigma^2} \log \Phi &= -\frac{n}{2\sigma^2} + \frac{1}{2} \frac{\sum x_j^2}{\sigma^4} \\ &= (t - \sigma^2)/\lambda(\sigma^2)\end{aligned}$$

provided that $\lambda = 2\sigma^4/n$, in which case $t = \sum x_j^2/n$. Again, $2\sigma^4/n$ is the variance of the estimate t of σ^2 .

If we had had to estimate both μ and σ^2 in

$$\phi(x; \mu, \sigma^2) \equiv (2\pi\sigma^2)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2}(x - \mu)^2/\sigma^2 \right\}$$

we should have proceeded by first estimating μ . The estimate proves as before to be the mean $\bar{x} = \sum x_j/n$, with variance $\lambda = \sigma^2/n$. This estimate being accepted, we know from standard results that the compound probability density *relative to the estimate* $\bar{x}$, not to the unknown μ , is

$$\Phi(x_1, x_2, \dots, x_n; \bar{x}, \sigma^2) = c(\sigma^2)^{-\frac{1}{2}(n-1)} \exp \left\{ -\frac{1}{2} \sum (x_j - \bar{x})^2/\sigma^2 \right\}.$$

From this, estimating anew for σ^2 and writing the customary s^2 instead of t , we find

$$s^2 = \sum (x_j - \bar{x})^2/(n-1) \quad \text{and} \quad \lambda = 2\sigma^4/(n-1),$$

the latter being the well-known sampling variance of s^2 . The sampling distribution of s^2 is known to be not normal, but of Gamma Type.

As a further point of interest, let us try to estimate, not σ^2 but σ in

$$\phi(x; \sigma) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp -\frac{1}{2}x^2/\sigma^2.$$

We obtain

$$\frac{\partial}{\partial \sigma} \log \Phi = -n/\sigma + \sum x_j^2/\sigma^3.$$

Evidently this cannot be put in the form $(t - \sigma)/\lambda(\sigma)$, where t is independent of σ , and λ is independent of the x_j . In this example, therefore, we may not estimate for σ , but must estimate for σ^2 . The reason is that the property of minimum variance is not preserved under non-linear transformations of the parameter. It follows that in most cases we must estimate, not θ but some function $\tau(\theta)$. What function τ is will appear in the sequel.

Example 3.—The Cauchy distribution

$$\phi(x; \mu) = \{1 + (x - \mu)^2\}^{-1}/\pi.$$

Let us try to estimate μ , the abscissa of symmetry. We have

$$\frac{\partial}{\partial \mu} \log \Phi = 2 \sum (x_j - \mu)/\{1 + (x_j - \mu)^2\}.$$

This cannot be put into the form $(t - \theta)/\lambda(\theta)$. The Cauchy distribution is thus intractable by the present method.

Example 4.—

$$\begin{aligned}\phi(x; \theta) &= \{\Gamma(k)\}^{-1} \theta^{-k} x^{k-1} e^{-x/\theta} \\ \frac{\partial}{\partial \theta} \log \Phi &= -nk\theta^{-1} + \sum x_j/\theta^2 \\ &= (t - \theta)/\lambda(\theta)\end{aligned}$$

provided that $t = \sum x_j/nk$, and $\lambda = \theta^2/nk$.

Here let us observe that the sample is finite and that the distribution of t is non-normal.

4. COMPARISON WITH MAXIMUM LIKELIHOOD.

It appears, then, that an estimating function $\tau(\theta)$ exists when $\partial \log \Phi / \partial \theta$ resolves itself into the form

$$\frac{\partial}{\partial \theta} \log \Phi = \frac{\partial \tau(\theta)}{\partial \theta} \{t(x_1, x_2, \dots, x_n) - \tau(\theta)\} / \lambda(\theta). \quad (1)$$

We estimate $\tau(\theta)$ by t , and deduce the estimate of θ . Thus our procedure, when possible, consists in virtually equating $t(x_1, x_2, \dots, x_n)$ to $\tau(\theta)$, indeed in formally solving the equation in θ ,

$$\frac{\partial}{\partial \theta} \log \Phi = 0. \quad (2)$$

Now this is precisely what is done in R. A. Fisher's method (Fisher, 1921) of maximum likelihood, for $\log \Phi$ is the logarithm of the compound probability density of the sample; but in that case it is done without consideration of the existence or nature of $\lambda(\theta)$, a question which does not arise. From our standpoint, on the other hand, the existence of $\lambda(\theta)$ is fundamental; for when $\lambda(\theta)$ exists, the estimate by maximum likelihood (of $\tau(\theta)$, that is, or the trivial linear function $a\tau(\theta) + b$, but not of non-linear functions of $\tau(\theta)$) has also minimum variance, even in the case of a finite sample.

5. VARIANCE OF ESTIMATE.

The minimal variance v of $\tau(\theta)$, that is the sampling variance of the estimate, can be expressed in several different ways. To avoid prolixity, let us suppose that the suitable parameter has been ascertained, and is henceforth called θ . We have then

$$v = \iiint \dots (t - \theta)^2 \Phi dx_1 dx_2 \dots dx_n, \quad (1)$$

where

$$t - \theta = \lambda(\theta) \frac{\partial}{\partial \theta} \log \Phi.$$

Hence

$$v = \lambda^2 \iiint \dots (\Phi_\theta / \Phi)^2 \Phi dx_1 dx_2 \dots dx_n. \quad (2)$$

Now

$$\frac{\partial^2}{\partial \theta^2} \log \Phi = \Phi^{-1} \frac{\partial^2 \Phi}{\partial \theta^2} - (\Phi_\theta / \Phi)^2$$

whence

$$(\Phi_\theta / \Phi)^2 = \frac{\partial^2 \Phi}{\partial \theta^2} \Phi^{-1} - \left(\frac{\partial^2}{\partial \theta^2} \log \Phi \right).$$

Hence, since the integrals of Φ_θ and $\Phi_{\theta\theta}$, the range being independent of θ , are both zero, we have

$$v = -\lambda^2 \iiint \dots \left(\frac{\partial^2}{\partial \theta^2} \log \Phi \right) \Phi dx_1 dx_2 \dots dx_n. \quad (3)$$

But again, since

$$\frac{\partial}{\partial \theta} \log \Phi = (t - \theta) / \lambda(\theta), \quad \frac{\partial^2}{\partial \theta^2} \log \Phi = (t - \theta) \frac{\partial}{\partial \theta} \{ \lambda(\theta) \}^{-1} - \{ \lambda(\theta) \}^{-1},$$

we have

$$v = -\lambda^2 \iiint \dots (-\lambda^{-1} \Phi) dx_1 dx_2 \dots dx_n, \quad (4)$$

since

$$\iiint \dots (t - \theta) \Phi dx_1 dx_2 \dots dx_n = 0.$$

Hence $v = \lambda$, while equally we see from (3) that v^{-1} is the mean value of $-\partial^2 L / \partial \theta^2$, where $L = \log \Phi$. This last is a well-known result of R. A. Fisher, which here we see holding accurately in finite samples, provided that $\lambda(\theta)$ exists, and that $\partial L / \partial \theta = (t - \theta) / \lambda(\theta)$.

It is easy to prove from $v = \lambda$ that the only Pearsonian distribution for which the mean is a sufficient statistic for locating the curve is the normal distribution. For since the variance of the mean of *any* distribution is σ^2/n , we must have

$$\begin{aligned} \frac{\partial}{\partial \theta} \log \Phi &= n(\theta - \sum x/n) / \sigma^2 \\ &= (n\theta - \sum x) / \sigma^2, \end{aligned}$$

whence

$$\frac{\partial}{\partial \theta} \log \phi = -(x - \theta) / \sigma^2$$

and so

$$\phi = c(x, \sigma) \exp \left\{ -\frac{1}{2}(x - \theta)^2 / \sigma^2 \right\}.$$

There is an arbitrary step here, namely, the choice of the arbitrary function c , which may involve x and σ but not θ , but a census of the Pearsonian curves will show that no other Pearsonian type except the normal will satisfy the problem. A probability function of Type A would, however, be admissible.

6. PROBABILITY FUNCTIONS AMENABLE TO THE POSTULATES.

By integrating the fundamental equation of estimation

$$\frac{\partial}{\partial \theta} \log \Phi = (t - \theta)/\lambda(\theta) \quad (1)$$

we may ascertain what class of probability function is amenable to the present method. Since t is independent of θ , and λ is independent of the x_j , we have

$$\log \Phi = \mu(\theta)(t - \theta) + \nu(\theta) + C, \quad (2)$$

where

$$\mu(\theta) = \int_0^\theta \frac{d\theta}{\lambda(\theta)}, \quad \nu(\theta) = - \int_0^\theta \frac{\theta d\theta}{\lambda(\theta)}, \quad (3)$$

and where C is a function of the x_j but not of θ . Thus $\phi(x; \theta)$ must be of the form

$$\phi(x; \theta) = \exp \{ \mu(\theta)t(x) + F(\theta) + f(x) \} \quad (4)$$

and must also satisfy the condition of total probability, § 3 (3). This class of functions includes the normal function and functions of Pearson's Type III.

By means of the above result we are able to determine what function $\tau(\theta)$ should be estimated, if necessary, instead of θ . For if

$$\phi = \exp (\mu t + F + f)$$

then

$$\Phi = \exp (\mu \sum t + nF + \sum f), \quad (5)$$

and so

$$\begin{aligned} \frac{\partial}{\partial \tau} \log \Phi &= \frac{\partial \mu}{\partial \tau} \sum t + n \frac{\partial F}{\partial \tau} \\ &= n \frac{\partial \mu}{\partial \tau} \left(\sum t/n + \frac{\partial F}{\partial \mu} \right). \end{aligned}$$

Hence if

$$\tau = -\partial F / \partial \mu = -\frac{\partial F}{\partial \theta} / \frac{\partial \mu}{\partial \theta},$$

we arrive at

$$\frac{\partial}{\partial \tau} \log \Phi = (\sum t/n - \tau) / \lambda(\tau). \quad (6)$$

Thus we estimate for $\tau(\theta)$, as given by $-\partial F/\partial\mu$, and our estimate is indeed $\sum t/n$, its variance being $(n\partial\mu/\partial\tau)^{-1}$.

Example.—

$$\phi(x; \sigma) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp -\frac{1}{2}x^2/\sigma^2.$$

Here

$$\mu = -1/2\sigma^2, \quad F = -\log \sigma,$$

whence

$$-\frac{\partial F}{\partial\mu} = -\frac{\partial F}{\partial\sigma} \frac{\partial\mu}{\partial\sigma} = \sigma^2,$$

which we must therefore estimate. Also $v = \left(n \frac{\partial\mu}{\partial\tau}\right)^{-1} = 2\sigma^4/n$.

Or again, purely in terms of μ and F , we might have written

$$\begin{aligned} v^{-1} &= n \frac{\partial\mu}{\partial\tau} = n \frac{\partial\mu}{\partial\theta} \frac{\partial\theta}{\partial\tau} \\ &= -n \frac{\partial\mu}{\partial\theta} \left/ \left(\frac{\partial^2 F}{\partial\mu^2} \cdot \frac{\partial\mu}{\partial\theta} \right) \right., \quad \left(\text{since } \tau = -\frac{\partial F}{\partial\mu} \right) \end{aligned}$$

whence

$$v = -n^{-1} \frac{\partial^2 F}{\partial\mu^2}.$$

7. SUFFICIENT STATISTICS AND MINIMUM VARIANCE.

In a paper, "On distributions admitting a sufficient statistic," B. O. Koopman finds the general form of a distribution function (Koopman, 1936) admitting the determination of sufficient statistics for the estimation of some or all of the parameters θ_j .

The intuitive notion of sufficiency is this, that a statistic should use up the whole of the relevant information contained in a sample. The formulation of this idea with respect to $\phi(x; \theta)$ leads to a sufficient statistic estimating θ , provided that the equality

$$\frac{\Phi(x_1, x_2, \dots, x_n; \theta)}{\Phi(x'_1, x'_2, \dots, x'_n; \theta)} = \frac{\Phi(x_1, x_2, \dots, x_n; \theta')}{\Phi(x'_1, x'_2, \dots, x'_n; \theta')}$$

is implied by $t(x'_1, x'_2, \dots, x'_n) = t(x_1, x_2, \dots, x_n)$ for any two values θ and θ' of the parameter and any two possible samples, as indicated.

Koopman finds that, if ϕ is analytic and non-zero over some continuous range of θ , then

$$\phi(x; \theta) = \exp \{F_1(\theta)f_1(x) + F_2(\theta) + f_2(x)\},$$

where the functions in the exponent are real, single-valued, analytic functions of their arguments.

Now this is the form which we obtained in § 6 for probability functions amenable to the postulates of unbiased estimate and minimum variance.

We see, therefore, that these functions also admit sufficient statistics for the estimation of the parameters concerned.

8. SIMULTANEOUS ESTIMATION OF PARAMETERS.

Problems of so-called *simultaneous* estimation lead to the following equations, in k parameters:

$$t_j(x_1, x_2, \dots, x_n) = \theta_j - \lambda_j(\theta_1, \theta_2, \dots, \theta_k) \frac{\partial}{\partial \theta_j} \log \Phi, \quad j = 1, 2, \dots, k,$$

where $t_1, t_2, \dots, t_k$ are functions of $x_1, x_2, \dots, x_n$ independent of $\theta_1, \theta_2, \dots, \theta_k$ and $\lambda_1, \lambda_2, \dots, \lambda_k$ are functions of $\theta_1, \theta_2, \dots, \theta_k$ independent of $x_1, x_2, \dots, x_n$. The strictly simultaneous solution of these equations is usually difficult. It is, indeed, extremely rare to find all the parameters admitting equations from which the factor $\lambda(\theta)$ is extricable. As we have seen, even the estimation of μ and σ^2 from a normal sample involves us in difficulty, unless we agree to estimate, first μ by $\bar{x}$, and then σ^2 from the corresponding modified probability density, relative to $\bar{x}$, not to μ . The estimation of the means μ_{10} and μ_{01} of x and y , the variances μ_{20} and μ_{02} and the correlation coefficient ρ from a bivariate normal sample of n paired values (x_i, y_i) offers still greater difficulties.

It is proposed to reserve the subject of simultaneous estimation and other aspects of the method of minimum variance for future consideration.

SUMMARY.

In the problem of estimating from sample the value of a parameter in a probability function new postulates are suggested of unbiased linear estimate and minimum sampling variance. A comparison is made, with illustrative examples, between this method and the principle of maximum likelihood, and ground common to the two is traversed. The new postulates are also placed in relation to the theory of sufficient statistics.

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